

A pattern is a conjecture

Name: _____

Date: _____

Two players alternate. Take **1, 3, or 4** counters each turn. Taking 2 is illegal. Never take more than remain. Taking the last counter wins.

1. Build the evidence.

Play from 10 and 14. Then label positions W or L from the viewpoint of the player about to move. Put L at zero.

0	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27

2. Make every label earn its place.

For W, give one legal move to L. For L, check all legal moves go to W. Find a disputed position and defend it.

What repeat do you conjecture? State it exactly.

Next question: Is seeing four repeats a proof that the pattern continues forever? What feature of the rules could force it?

Build first. Explain by pointing, talking, or drawing.

Prove the repeat forever

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3. Two obligations.

Your proposed losing piles must satisfy **both** tests: (a) no legal move goes from one proposed L to another; (b) every other positive pile has a legal move to a proposed L.

4. Use groups of seven.

Write a pile as some whole groups of seven plus a remainder from 0 to 6. Complete the table using your conjecture.

Remainder	W or L	Take this many	New remainder
0			
1			
2			
3			
4			
5			
6			

Why do the two obligations prove a winning strategy?

Check the boundary: Your suggested move must be legal even for the smallest positive pile in that remainder class. Why does a decreasing pile eventually reach zero?

Build first. Explain by pointing, talking, or drawing.

Change a rule; challenge a proof

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New game: take **1, 3, or 5**. Two players alternate. Taking the last counter wins. This is not “take any number from 1 to 5.”

5. Predict before playing.

Who can force a win from 6? From 7? From 100? Test small piles, then find one explanation for every pile.

My new losing piles and both parts of my proof:

6. How much memory does the old game need?

Return to moves 1, 3, 4. Copy a stretch of your labels onto a straight strip. Put a paper frame around **four consecutive W/L labels**. Which three cells decide the next label? Add that label, then slide the frame one cell right. Find a matching window elsewhere and show why both next labels must agree.

Why identical four-label windows have identical futures:

Go further: A four-label window has at most $2^4 = 16$ patterns. Explain why some window must recur. Does this prove repetition starts at zero, or only that it eventually starts?

Build first. Explain by pointing, talking, or drawing.